EXPERIMENT 9

Aim:

Create an ERC 20 Token in Remix IDE

THEORY

1. What is a Token?

A token is a digital asset created and managed using a blockchain's smart contract functionality. Tokens can
represent currencies, assets or utilities.

Common token standards include:

· ERC-20: Fungible tokens.
ERC-721: Non-fungible tokens (NFTs).
. ERC-1155: Supports both fungible and non-fungible tokens.

2. What is ERC-20?

ERC-20 (Ethereum Request for Comment 20) is a standard for creating fungible tokens on the Ethereum blockchain.
Each token is interchangeable with another token of the same type.

It defines common functions and events that allow wallets, exchanges and DApps to interact with tokens.

3. Remix IDE

Remix IDE is a browser-based development environment used to write, compile, deploy and test smart contracts
written in Solidity.

It provides tools for compiling contracts, connecting wallets and deploying smart contracts on blockchain networks.

4. MetaMask

MetaMask is a cryptocurrency wallet used to manage Ethereum accounts and interact with decentralized

applications. It connects Remix IDE to the blockchain and is used to approve transactions and pay gas fees.

5. Sepolia Testnet

Sepolia is an Ethereum test network used for developing and testing smart contracts. It uses test ETH instead of real
ETH to perform transactions and pay gas fees.

6. ERC-20 Token Functions

.Important ERC-20 functions include:

Function	Purpose
`name()`	Returns token name
`symbol()`	Returns token symbol
`decimals()`	Returns decimal precision
`totalSupply()`	Returns total token supply
`balanceOf()`	Checks account balance
`transfer()`	Transfers tokens
`approve()`	Approves token spending
`allowance()`	Checks approved spending limit
`transferFrom()`	Transfers approved tokens

7. ERC-20 Events

Events record important activities performed by a smart contract.

· Transfer: Emitted whenever tokens are transferred.
. Approval: Emitted whenever an owner approves another account to spend tokens.

8. ERC-20 Attributes

. Name: Full name of the token.
. Symbol: Short representation of the token.
. Decimals: Number of decimal places supported, commonly 18.
. Total Supply: Total number of tokens created.
. Balance: Number of tokens held by an account.

9. ERC-20 vs ERC-721

ERC-20	ERC-721
Fungible tokens	Non-fungible tokens
Tokens are interchangeable	Each token is unique
Used for currencies and utility tokens	Used for NFTs
No unique ID for each unit	Each token has a unique ID

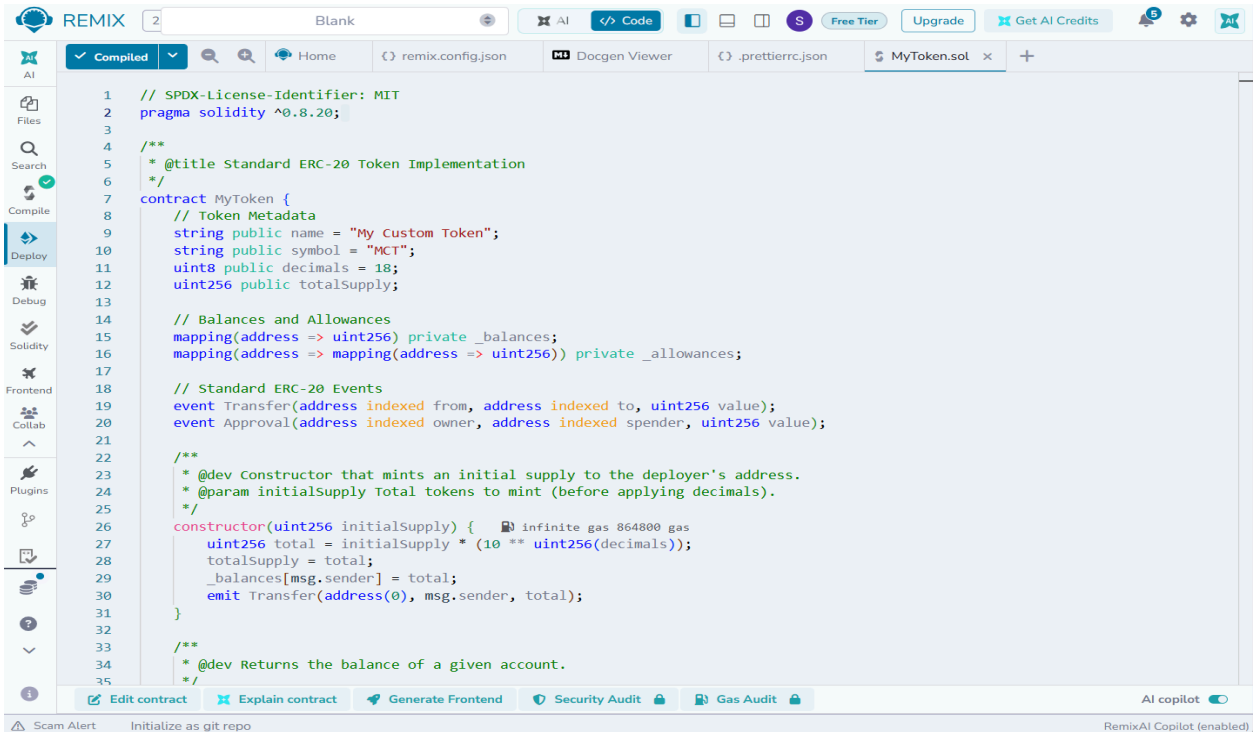
10. ERC-20 Extensions

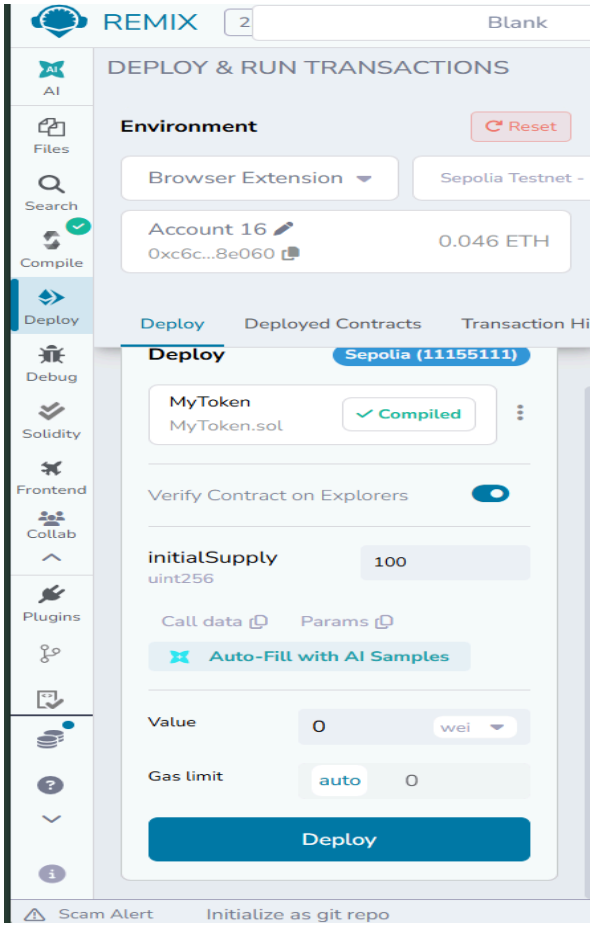
Mintable: Allows creation of new tokens, increasing the total supply.

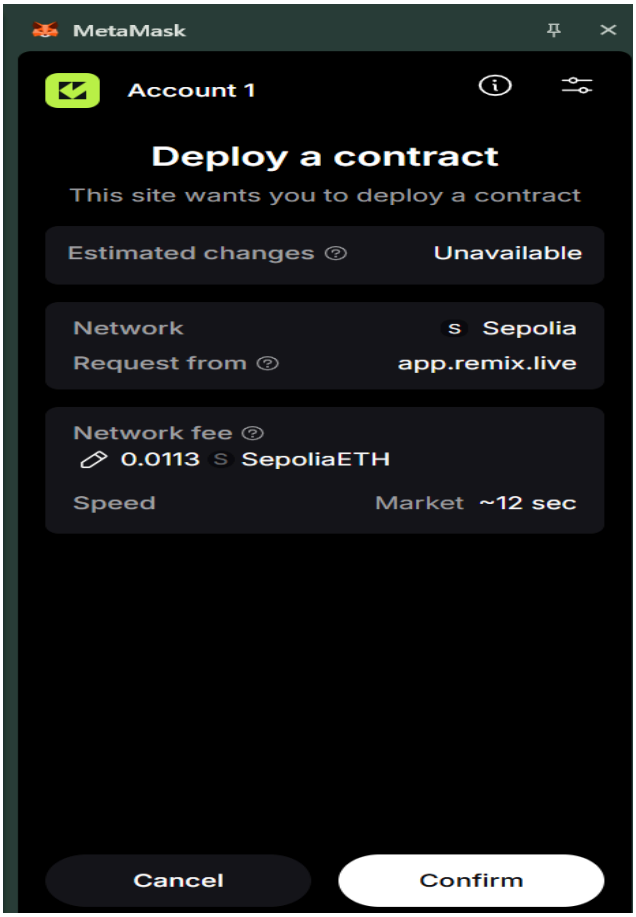
Burnable: Allows permanent destruction of tokens, decreasing the total supply.

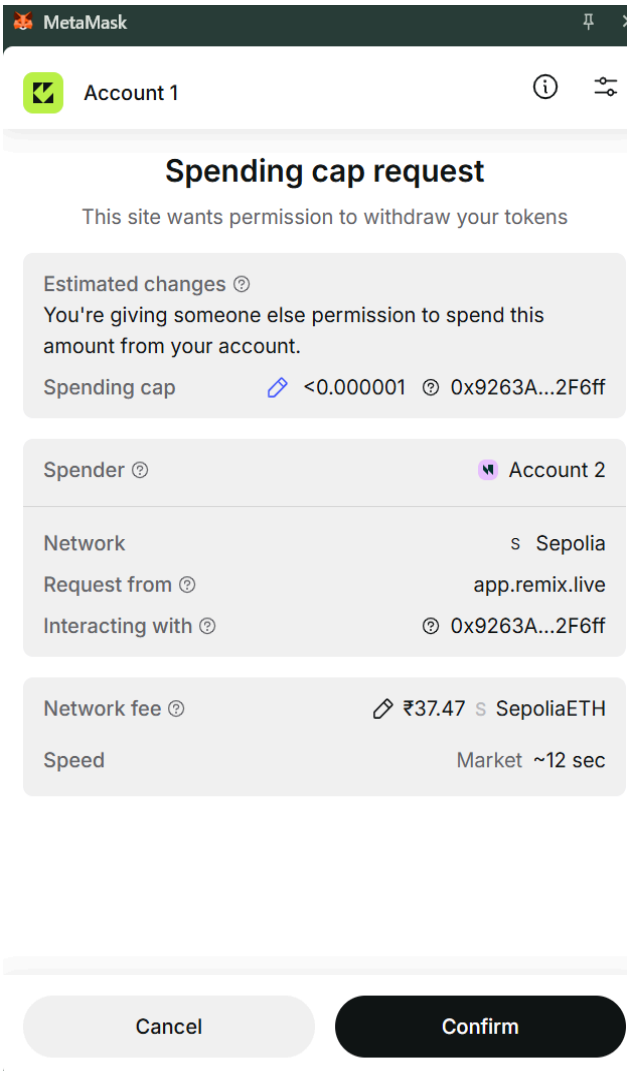
Pausable: Allows authorized accounts to temporarily stop token transfers during emergencies.

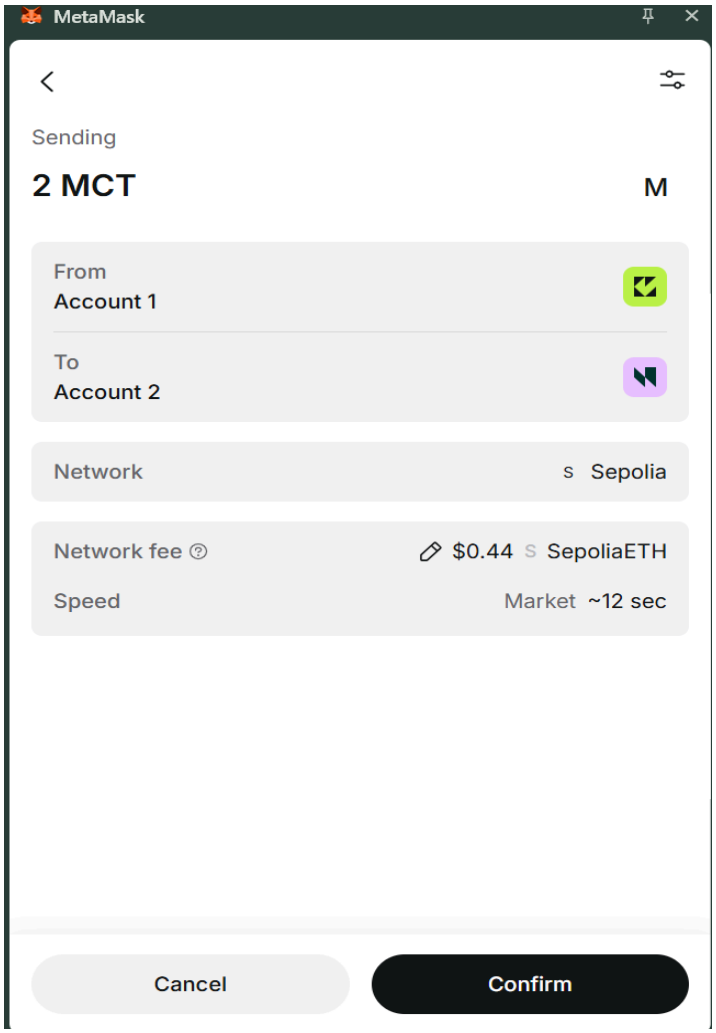
Output & Code:


REMIX
Blank
AI
Code
Free Tier
Upgrade
Get AI Credits
Compiled
Home
remix.config.json
Docgen Viewer
.prettierrc.json
MyToken.sol
AI
Files
Search
Compile
Deploy
Debug
Solidity
Frontend
Collab
Plugins
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
/**
* @title Standard ERC-20 Token Implementation
*/
contract MyToken {
// Token Metadata
string public name = "My Custom Token";
string public symbol = "MCT";
uint8 public decimals = 18;
uint256 public totalSupply;
// Balances and Allowances
mapping(address => uint256) private _balances;
mapping(address => mapping(address => uint256)) private _allowances;
// Standard ERC-20 Events
event Transfer(address indexed from, address indexed to, uint256 value);
event Approval(address indexed owner, address indexed spender, uint256 value);
/**
* @dev Constructor that mints an initial supply to the deployer's address.
* @param initialSupply Total tokens to mint (before applying decimals).
*/
constructor(uint256 initialSupply) {
infinite gas 864800 gas
uint256 total = initialSupply * (10 ** uint256(decimals));
totalSupply = total;
_balances[msg.sender] = total;
emit Transfer(address(0), msg.sender, total);
}
/**
* @dev Returns the balance of a given account.
*/
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)


REMIX
2
Blank
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
Sepolia Testnet -
Account 16
0xc6c...8e060
0.046 ETH
Deploy
Deployed Contracts
Transaction Hi
Deploy
Sepolia (11155111)
MyToken
MyToken.sol
Compiled
Verify Contract on Explorers
initialSupply
uint256
100
Call data
Params
Auto-Fill with AI Samples
Value
0
wei
Gas limit
auto
0
Deploy
Scam Alert
Initialize as git repo


MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
Unavailable
Network
Sepolia
Request from
app.remix.live
Network fee
0.0113 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm





Conclusion

Thus, the ERC-20 token was successfully created using Solidity in Remix IDE and deployed on the
Sepolia Testnet using MetaMask. The token was assigned an initial supply of 1,000,000 College Tokens
(CLG) with 18 decimals. The experiment demonstrated token creation, deployment, approval and transfer
operations, providing practical understanding of ERC-20 standards and smart contract interactions.